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Matsumoto

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(54) **UNIT LOCATING MECHANISM AND IMAGE FORMING APPARATUS**

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(22) Filed: **Aug. 20, 2024**

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(51) **Int. Cl.**

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G03G 15/01	(2006.01)
G03G 21/16	(2006.01)
G03G 21/18	(2006.01)

(52) **U.S. Cl.**

CPC ***G03G 15/1685*** (2013.01); ***G03G 15/0131*** (2013.01); ***G03G 15/0189*** (2013.01); ***G03G 15/1605*** (2013.01); ***G03G 21/1619*** (2013.01); ***G03G 21/1623*** (2013.01); ***G03G 21/1628*** (2013.01); ***G03G 21/1633*** (2013.01); ***G03G 21/1647*** (2013.01); ***G03G 21/168*** (2013.01); *G03G 21/1842* (2013.01)

(58) **Field of Classification Search**

CPC G03G 15/1685; G03G 15/0131; G03G 15/0189; G03G 15/1605; G03G 21/1619; G03G 21/1623; G03G 21/1628; G03G 21/1633; G03G 21/1647; G03G 21/168; G03G 21/1842

See application file for complete search history.

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(57) **ABSTRACT**

A unit locating mechanism includes a first unit, a second unit and a locating member. The locating member includes a cylindrical portion, a hook portion and a portion to be restricted. The second unit includes a unit-side portion, a support shaft and a restriction portion. The restriction portion overlaps, in a state where the hook portion engages with a boss portion, the portion to be restricted from outside in a predetermined direction. The unit locating mechanism includes a torsion coil spring. A first arm portion engages with the locating member, and a second arm portion engages with the unit-side portion such that the torsion coil spring biases the locating member in a direction which causes the hook portion to engage with the boss portion.

4 Claims, 13 Drawing Sheets

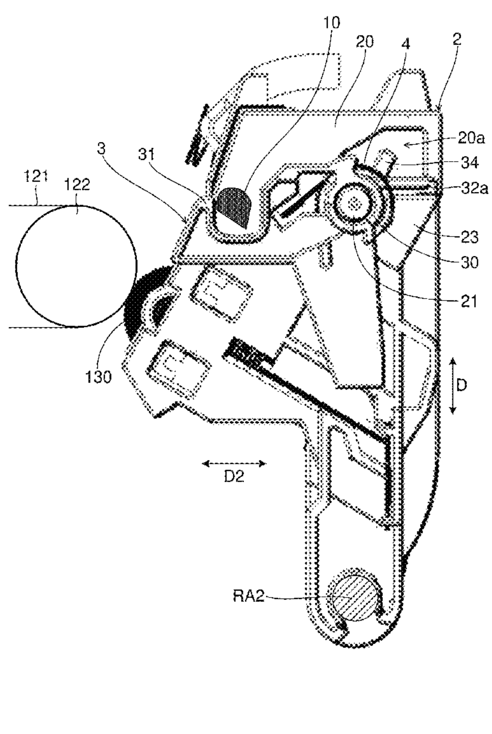


FIG.1

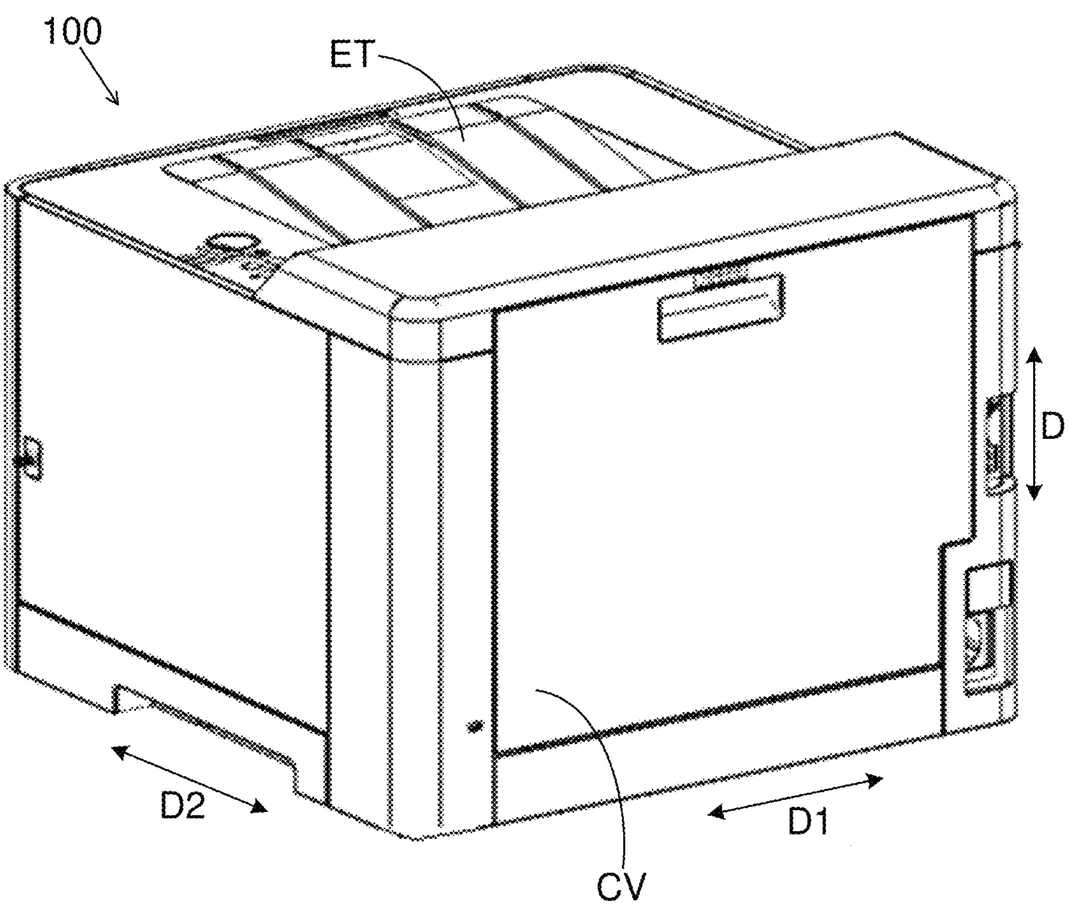


FIG.2

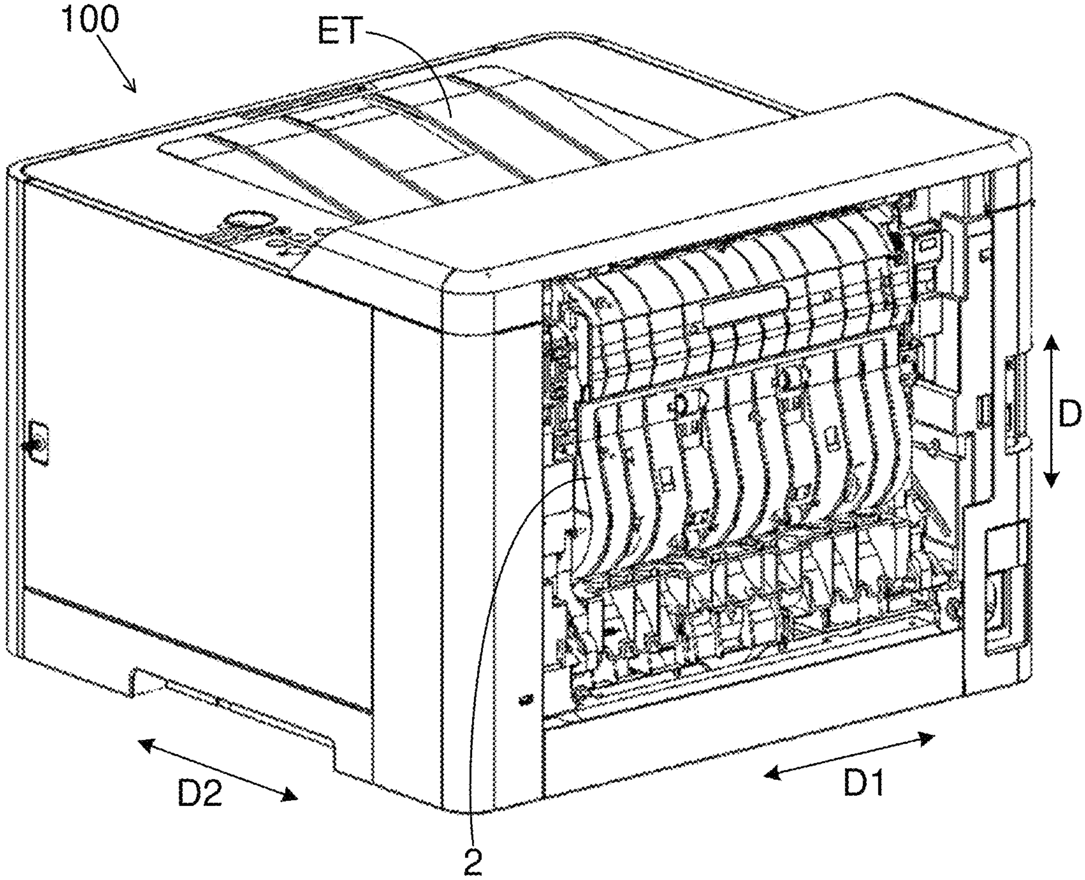
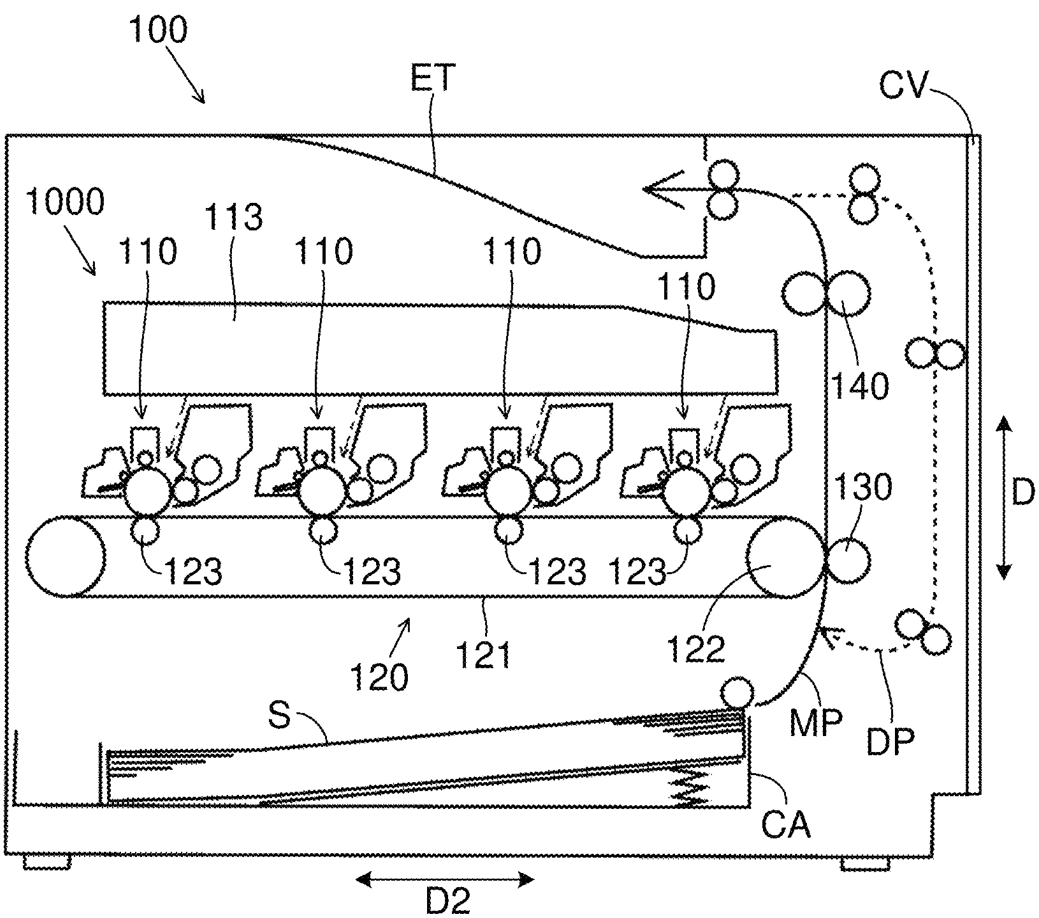


FIG.3


100
ET
CV
1000
113
110
110
110
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140
130
D
123
123
123
123
122
121
120
MP
DP
S
CA
D2

FIG.4

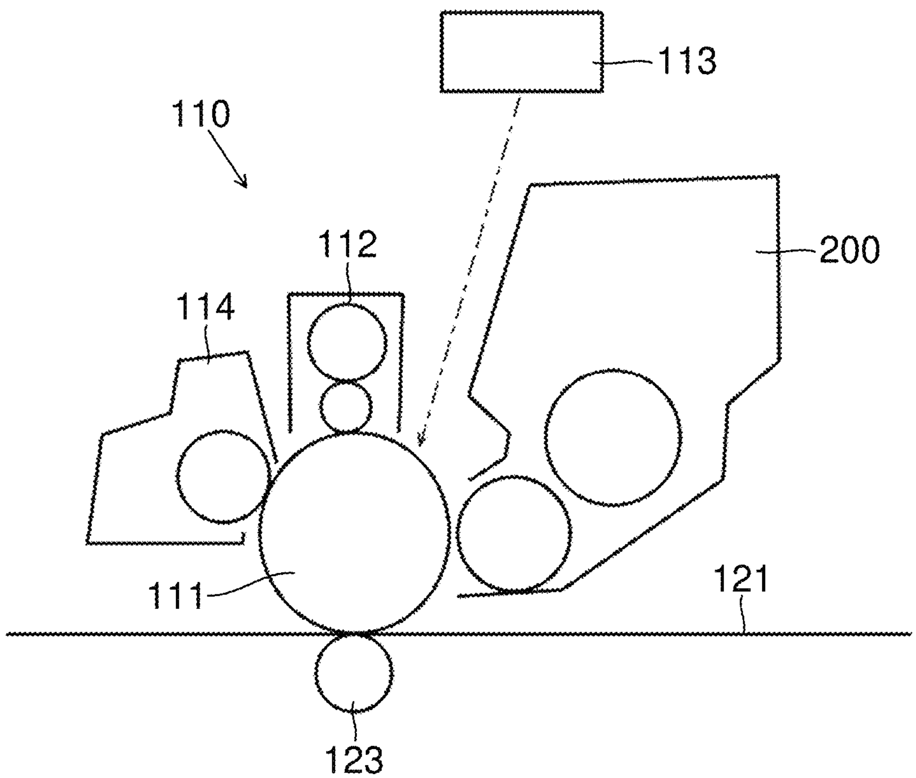


FIG.5

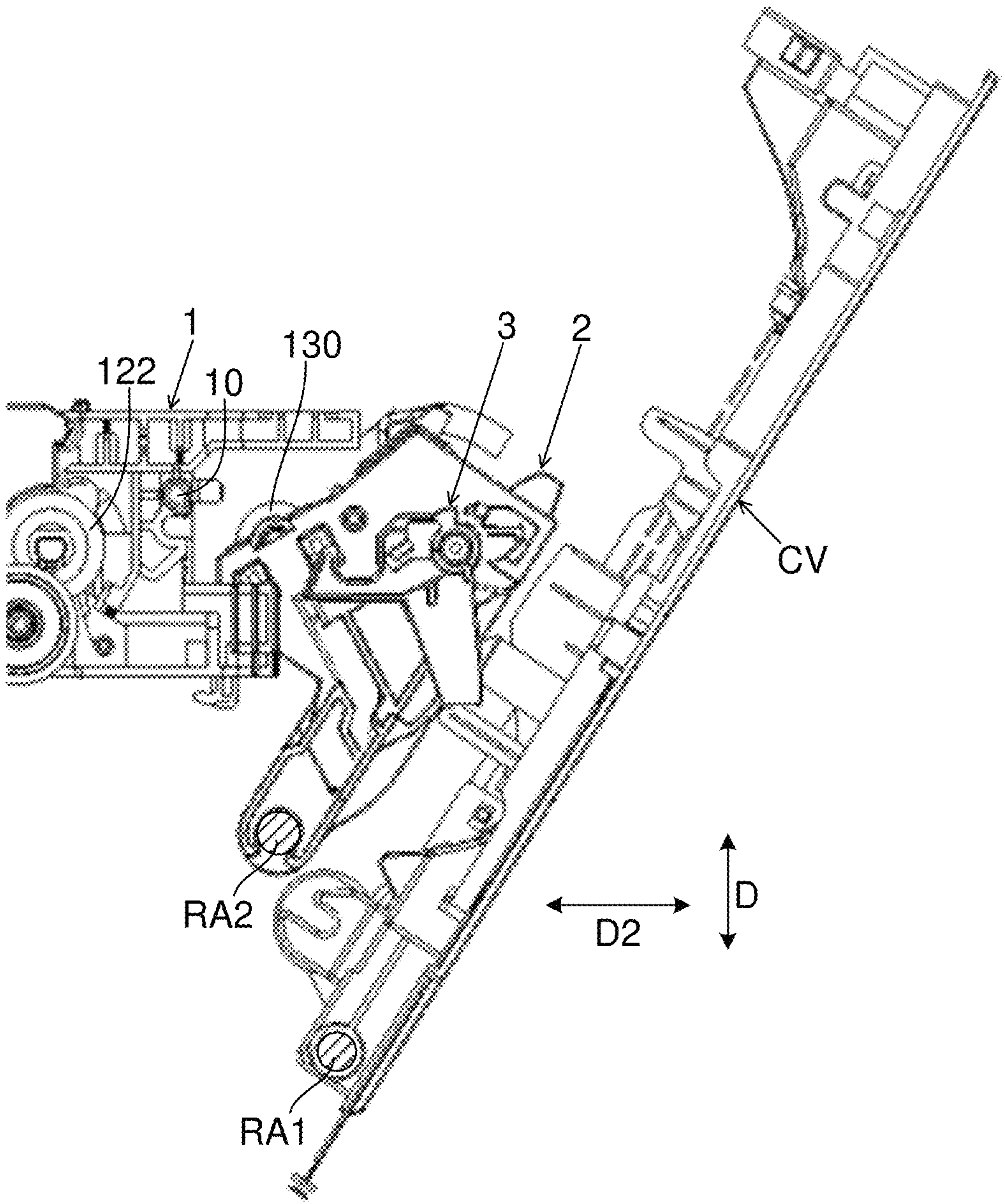


FIG.6

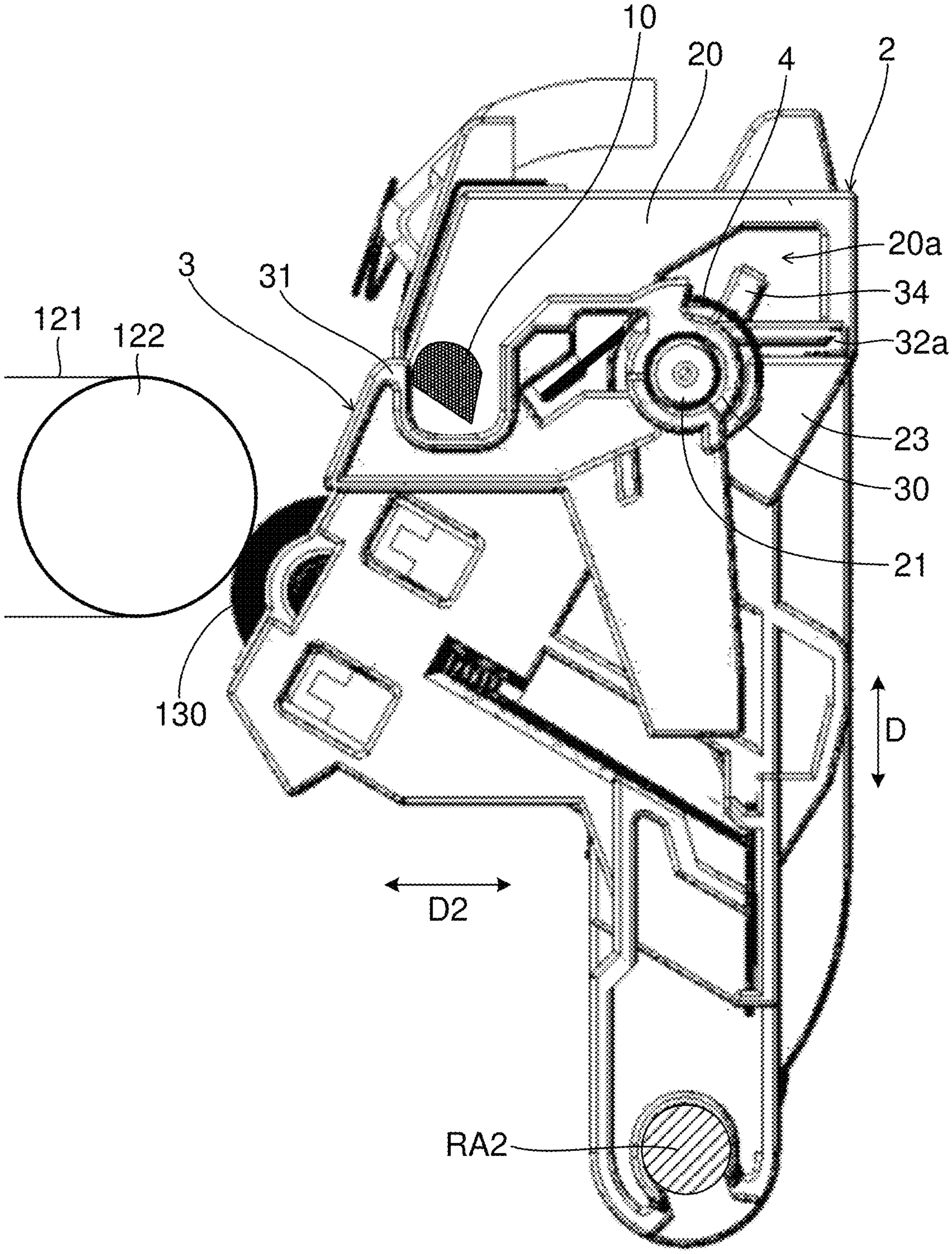
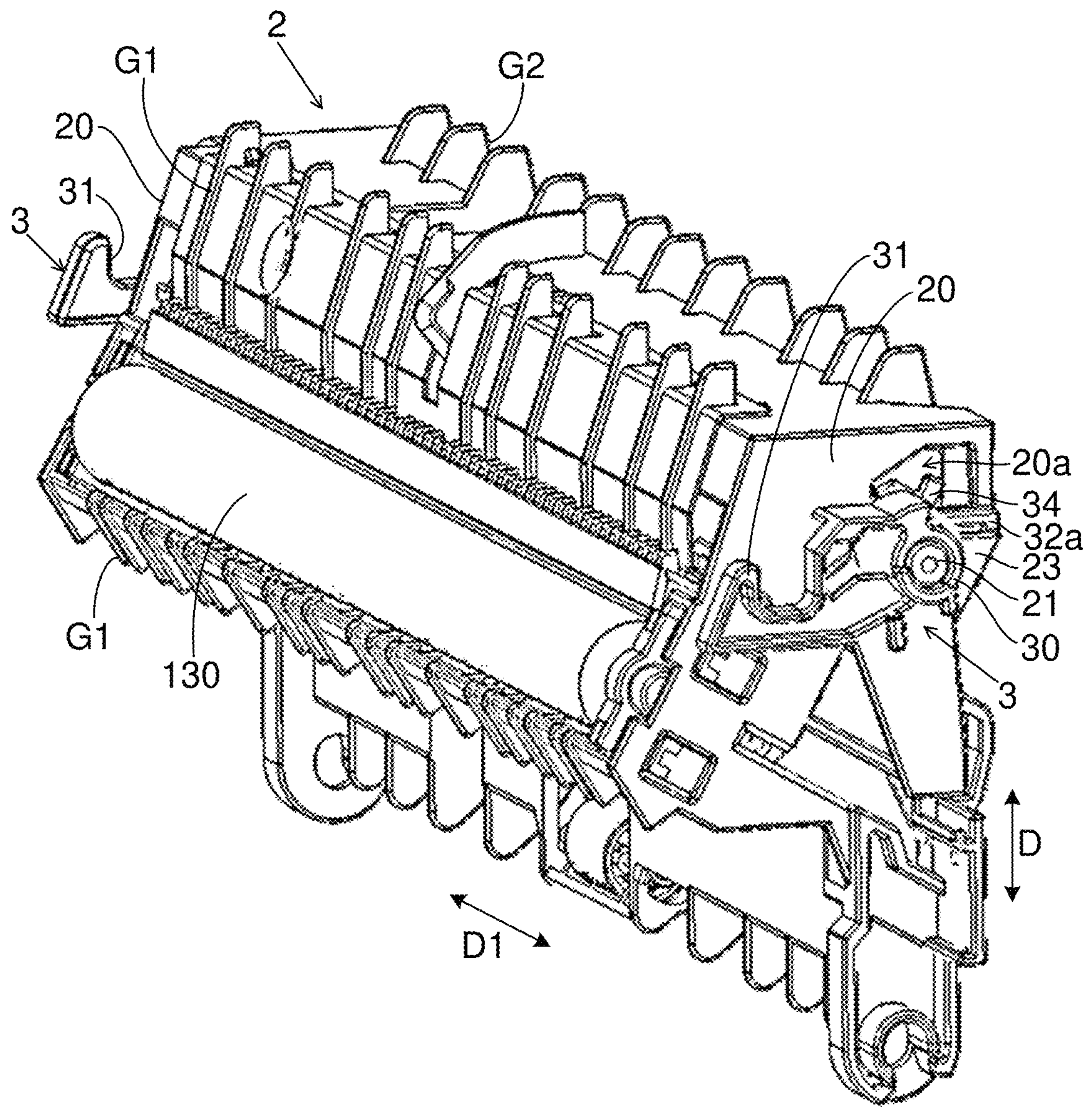


FIG.7



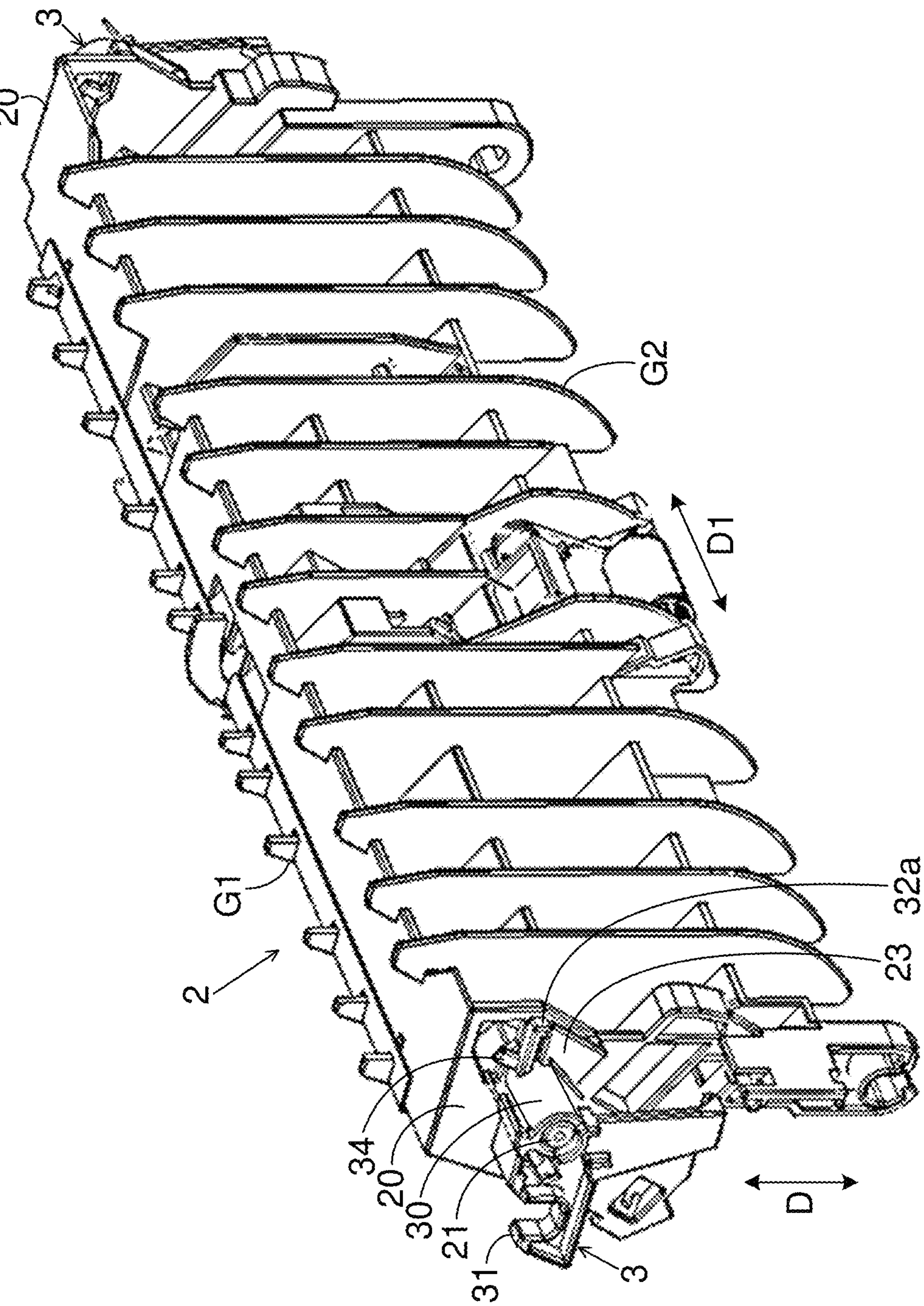


FIG.8

FIG.9

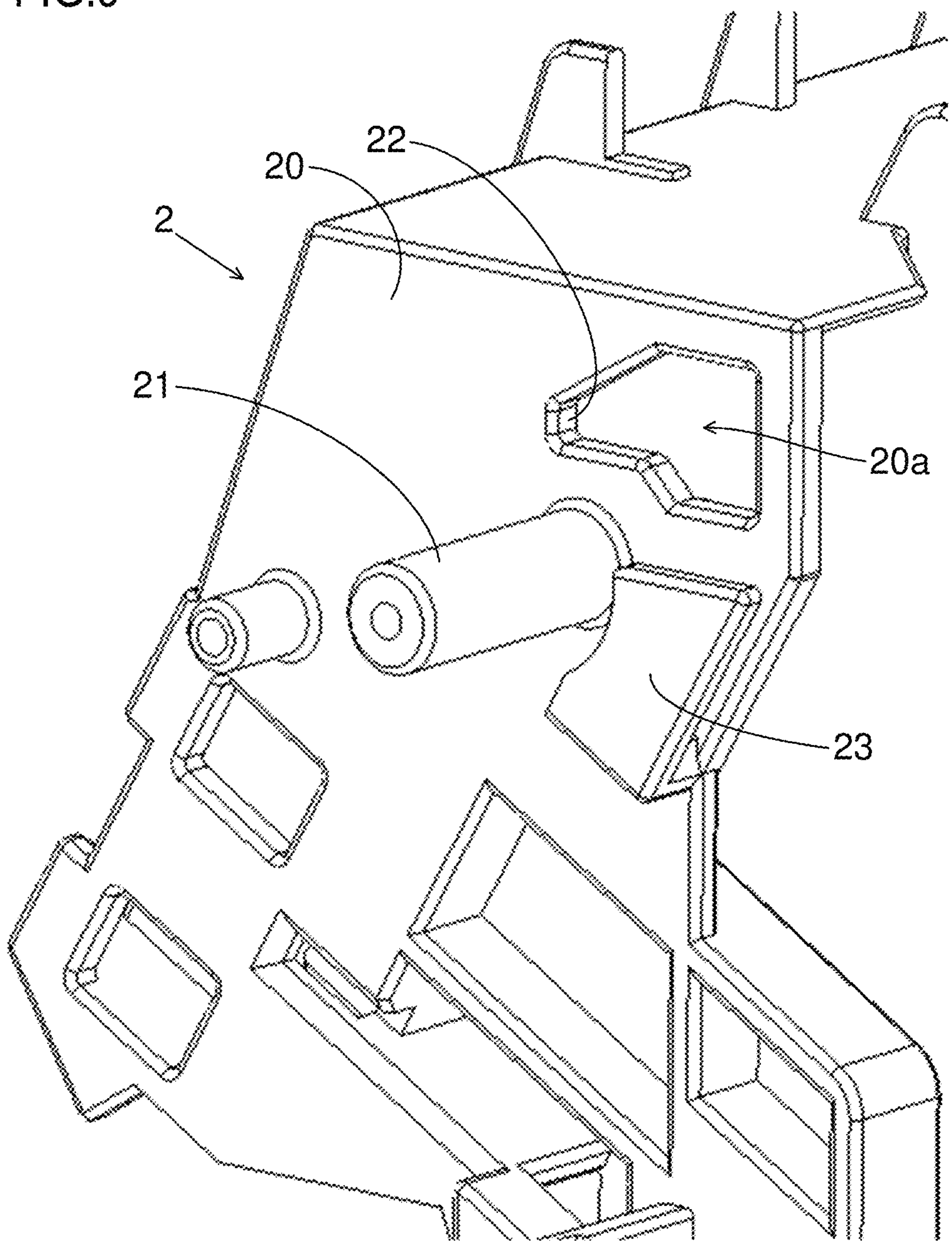


FIG.10

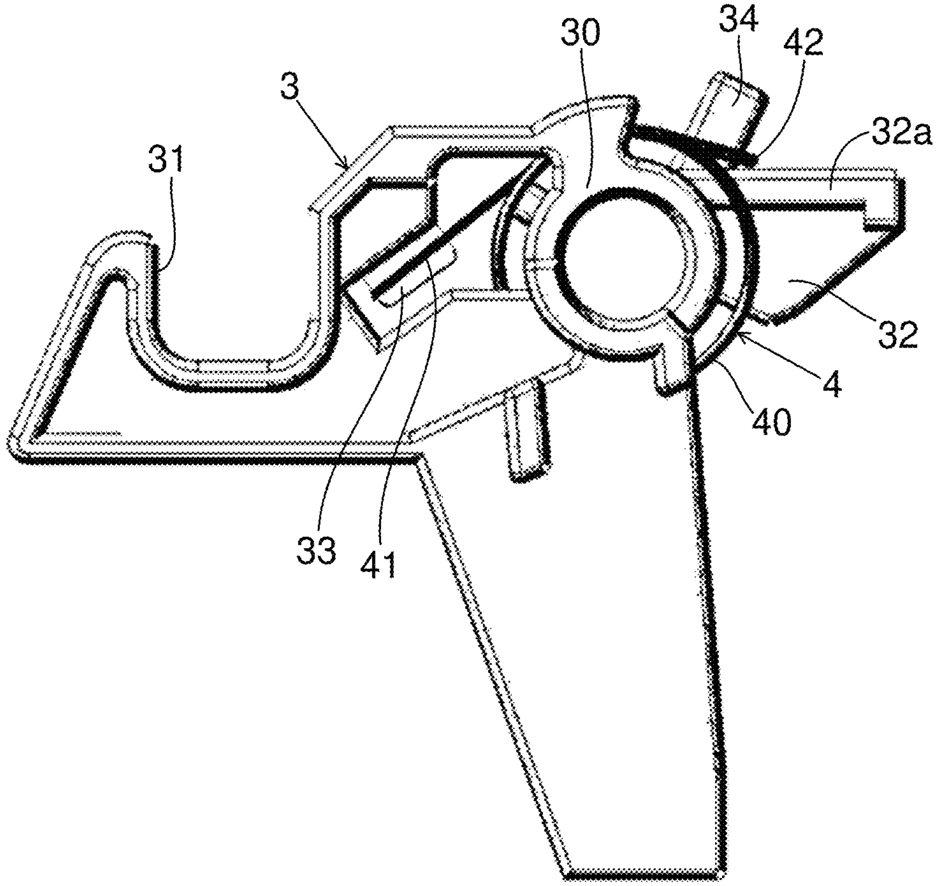


FIG.11

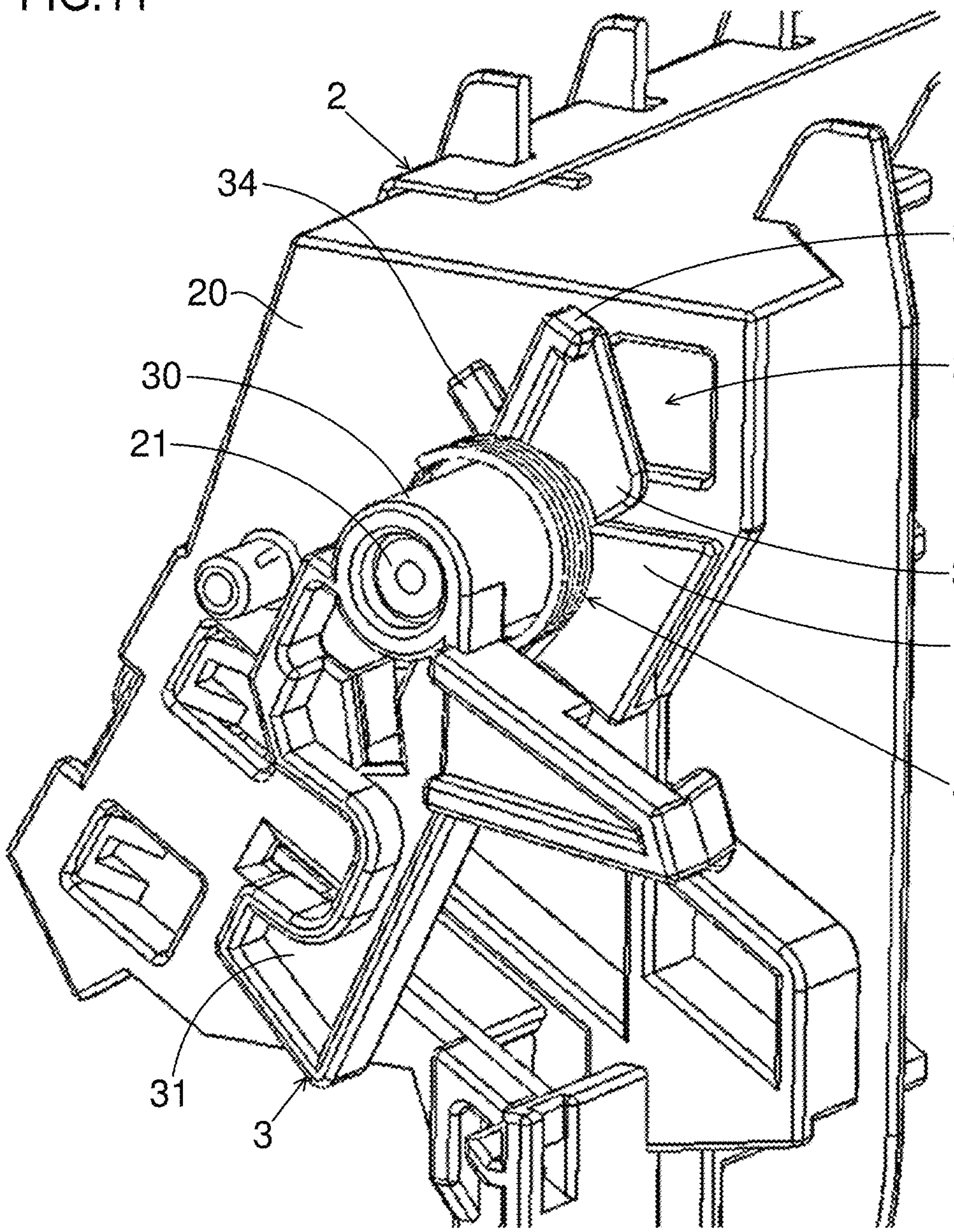


FIG.12

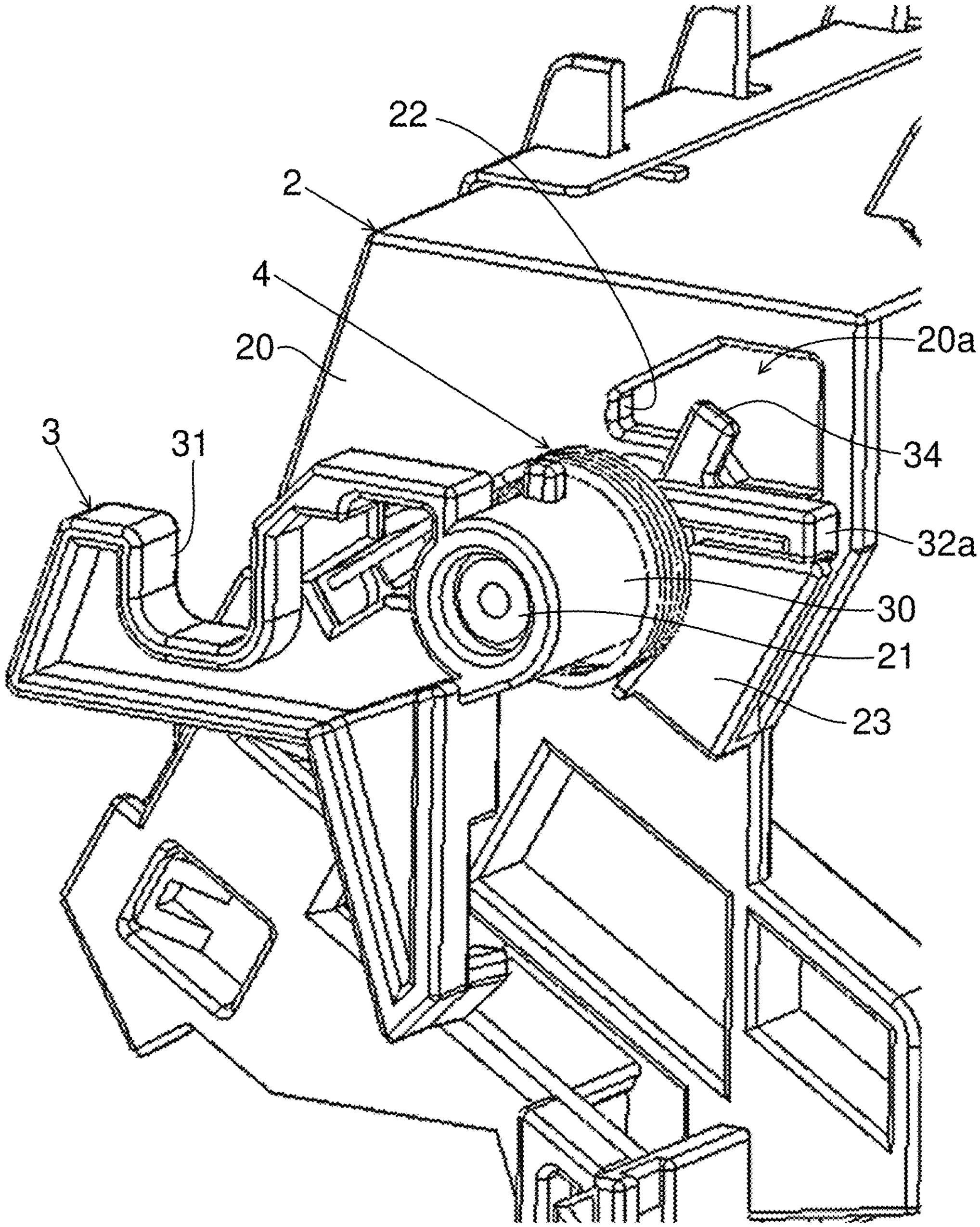
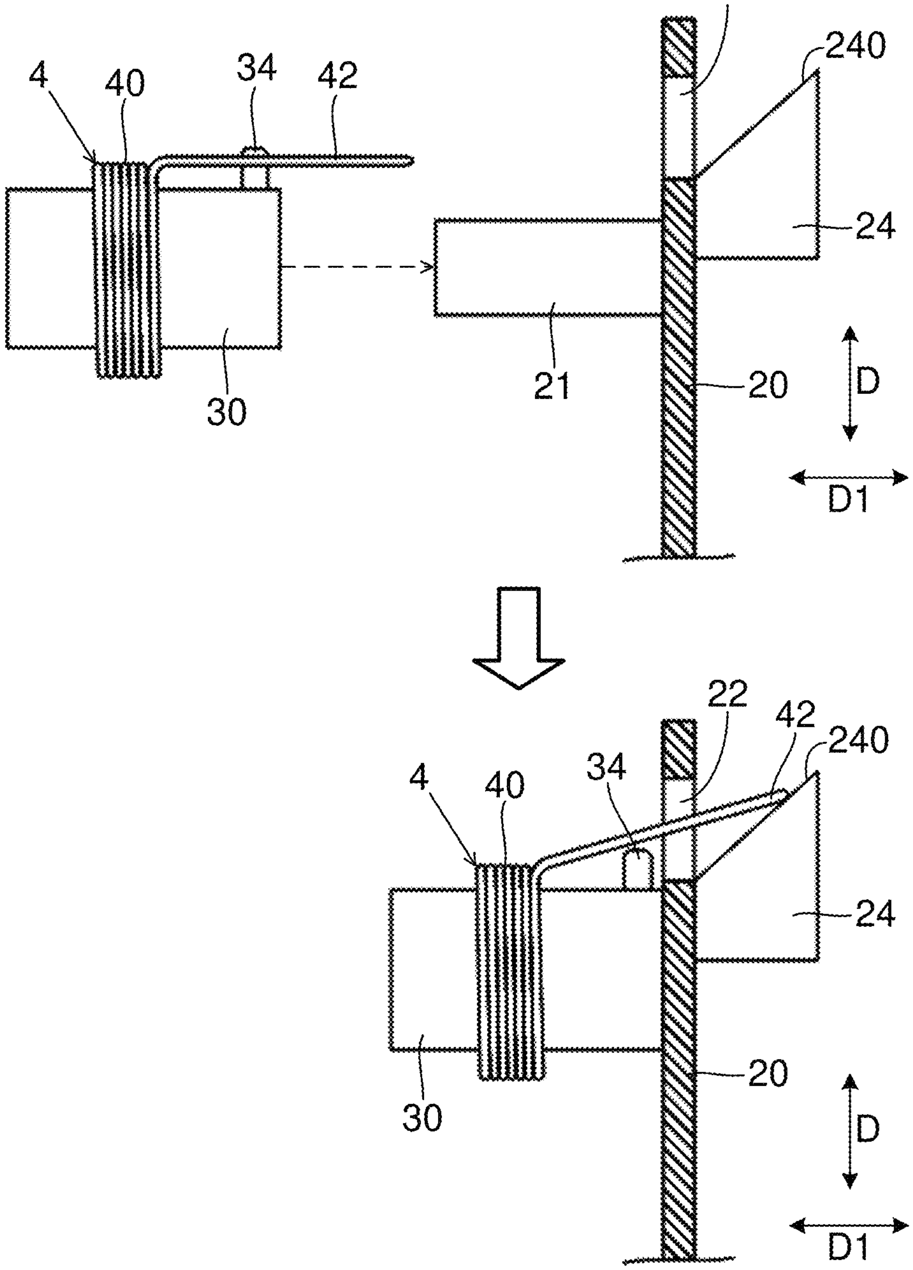


FIG.13



UNIT LOCATING MECHANISM AND IMAGE FORMING APPARATUS

INCORPORATION BY REFERENCE

This application is based upon and claims the benefit of priority from the corresponding Japanese Patent Application No. 2023-135531, filed on Aug. 23, 2023, the entire contents of which are incorporated herein by reference.

BACKGROUND

The present disclosure relates to unit locating mechanisms and image forming apparatuses.

A conventional image forming apparatus includes a unit which can be turned with one end edge used as a support point. For example, the unit is conventionally a cover member.

Conventionally, for example, the hook portion of a locating member engages with a unit to restrict the turning of the unit. In other words, the unit is located. When in this configuration, the locating member is shifted (for example, the locating member drops off), the unit is also shifted. However, when the locating member is fixed so as not to be shifted, it is difficult to disassemble the locating member and an area therearound.

SUMMARY

A unit locating mechanism according to a first aspect of the present disclosure includes a first unit, a second unit and a locating member. The first unit includes a boss portion. The second unit can be displaced between a first position located with respect to the first unit and a second position other than the first position by being turned around an axis line extending in a predetermined direction. The locating member retains the second unit in the first position. The locating member includes a cylindrical portion, a hook portion and a portion to be restricted. In the cylindrical portion, the predetermined direction is the direction of a cylindrical shaft. The hook portion is arranged radially outward of the cylindrical portion when viewed in the predetermined direction and engages with the boss portion so as to hold the boss portion from one side in a circumferential direction around the center of the cylindrical shaft of the cylindrical portion. The portion to be restricted is arranged radially outward of the cylindrical portion and in a position different from the hook portion in the circumferential direction when viewed in the predetermined direction. The second unit includes a unit-side portion, a support shaft and a restriction portion. The support shaft protrudes outward in the predetermined direction from the unit-side portion, the cylindrical portion is fitted from outside in the predetermined direction and the support shaft supports the locating member such that the locating member can be displaced in the circumferential direction. The restriction portion is arranged on the turning path of the portion to be restricted which is turned around the center of the cylindrical shaft of the cylindrical portion when viewed in the predetermined direction. The hook portion engages with the boss portion when the second unit is in the first position to restrict the turning of the second unit from the first position to the second position. The restriction portion overlaps, in a state where the hook portion engages with the boss portion, the portion to be restricted from the outside in the predetermined direction so as to restrict movement of the locating member outward in the predetermined direction. The unit locating

mechanism further includes a torsion coil spring that includes a winding spring portion which is arranged on the cylindrical portion, a first arm portion which extends from one end of the winding spring portion and a second arm portion which extends from the other end of the winding spring portion. The first arm portion engages with the locating member, and the second arm portion engages with the unit-side portion such that the torsion coil spring biases the locating member in a direction which causes the hook portion to engage with the boss portion.

An image forming apparatus according to a second aspect of the present disclosure includes the unit locating mechanism described above. The first unit is a unit that includes an image carrying member which carries an image. The second unit is a unit that includes a transfer roller which is pressed against the image carrying member. The image forming apparatus conveys a sheet into a transfer nip formed between the image carrying member and the transfer roller, and the image is transferred to the sheet.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a perspective view of an image forming apparatus according to an embodiment;

FIG. **2** is a perspective view of the image forming apparatus shown in FIG. **1** in a state where a rear cover is removed;

FIG. **3** is a schematic view showing an internal configuration of the image forming apparatus according to the embodiment;

FIG. **4** is a schematic view showing the configuration of an image formation unit in the image forming apparatus according to the embodiment;

FIG. **5** is a side view in a state where a conveyance unit is separate from a main body-side unit in the image forming apparatus according to the embodiment;

FIG. **6** is a side view of the conveyance unit and a locating member attached to the conveyance unit in the image forming apparatus according to the embodiment;

FIG. **7** is a perspective view when the conveyance unit and the locating member attached to the conveyance unit in the image forming apparatus according to the embodiment are viewed from the front;

FIG. **8** is a perspective view when the conveyance unit and the locating member attached to the conveyance unit in the image forming apparatus according to the embodiment are viewed from the back;

FIG. **9** is an enlarged perspective view of an area around the unit-side portion of the conveyance unit in the image forming apparatus according to the embodiment;

FIG. **10** is a side view of the locating member and a torsion coil spring temporarily fixed to the locating member in the image forming apparatus according to the embodiment;

FIG. **11** is a perspective view for illustrating a step of attaching the locating member in the embodiment;

FIG. **12** is a perspective view in a state where the locating member shown in FIG. **11** is turned to one side in a circumferential direction; and

FIG. **13** is a schematic view for illustrating a step of attaching a locating member in a variation.

DETAILED DESCRIPTION

An image forming apparatus **100** according to an embodiment of the present disclosure and an image forming apparatus **100** according to a variation of the embodiment will be described below with reference to FIGS. **1** to **13**. Although in the following description, a color laser printer which can perform a print job for printing an image on a sheet S is used as an example, the present disclosure is not limited to a printer, and can be applied to a multi-functional peripheral which has a copying function and the like.

In the following description, a direction perpendicular to a flat floor surface on which the image forming apparatus **100** is installed is defined as an up/down direction, and the up/down direction is identified with a symbol D. Of horizontal directions orthogonal to the up/down direction D, one direction is identified with a symbol D**1**, and the other direction orthogonal to the one direction D**1** is identified with a symbol D**2**.

For example, the one direction D**1** is the width direction (left/right direction) of the image forming apparatus **100**. In the following description, the one direction D**1** is referred to as the width direction D**1**. The width direction D**1** corresponds to a "predetermined direction". The other direction D**2** is the forward/backward direction of the image forming apparatus **100**. In the following description, the other direction D**2** is referred to as the forward/backward direction D**2**.

<Overall Configuration of Image Forming Apparatus>

The image forming apparatus **100** of the present embodiment has an appearance as shown in FIGS. **1** and **2**. The image forming apparatus **100** includes a main body frame (symbol of which is omitted). The image forming apparatus **100** also includes a rear cover CV. The main body frame is covered with exterior covers including the rear cover CV (for the other covers, symbols are omitted). In FIG. **2**, the rear cover CV is omitted, and a part of the interior of the apparatus is shown.

The main body frame supports the exterior covers. The main body frame supports the rear cover CV such that the rear cover CV can be turned around an axis line extending in the width direction D**1**. For example, the main body frame includes a turning shaft RA**1** (see FIG. **5**) which extends in the width direction D**1**. The turning shaft RA**1** supports an end portion of the rear cover CV on a lower side. The rear cover CV is turned with the turning shaft RA**1** used as a support point such that an end portion (that is, an end portion on an upper side) on a side opposite to the side supported by the turning shaft RA**1** swings. In other words, the rear cover CV is supported such that the rear cover CV can be opened or closed with respect to the main body frame.

The rear cover CV is opened, and thus the interior of the image forming apparatus **100** is exposed to one side (backward side) in the forward/backward direction D**2**. The rear cover CV is closed, and thus the interior of the image forming apparatus **100** is covered from the one side in the forward/backward direction D**2**. For example, when a jam of a sheet S occurs in the image forming apparatus **100**, the rear cover CV is opened so that the sheet S is removed.

As shown in FIG. **3**, the image forming apparatus **100** includes a main conveyance path MP. The main conveyance path MP is passed through a transfer nip which will be described later. In FIG. **3**, the main conveyance path MP is schematically indicated by a solid line with an arrow. A double-sided printing conveyance path DP which will be described later is schematically indicated by a dotted line with an arrow.

The image forming apparatus **100** includes a sheet cassette CA. The sheet cassette CA stores sheets S used in a print job. The type of sheet S is not particularly limited.

In the print job, the image forming apparatus **100** supplies the sheet S in the sheet cassette CA to the main conveyance path MP, and conveys the sheet S along the main conveyance path MP. Then, the image forming apparatus **100** prints an image on the sheet S being conveyed. In other words, the image forming apparatus **100** includes a print unit **1000** which prints the image on the sheet S being conveyed.

The print unit **1000** includes image formation units **110** for four colors of cyan, magenta, yellow and black. The image formation units **110** form toner images for the corresponding colors, respectively. With attention focused on a certain image formation unit **110**, the configuration of the image formation unit **110** will be described. The image formation units **110** basically have the same configuration. Hence, the description of the configuration of the other image formation units **110** is omitted because the following description can be used for the description.

As shown in FIG. **4**, the image formation unit **110** includes a development cartridge **200**. The development cartridge **200** stores the toner of the corresponding color, and uses the toner to perform a development process. The image formation unit **110** includes a photosensitive drum **111** and a charging device **112**. The image forming apparatus **100** includes one exposure unit **113** for the image formation units **110** of the four colors. The image formation unit **110** includes a cleaning device **114**.

When image formation is performed by the image formation unit **110**, the photosensitive drum **111** is rotated. The charging device **112** charges the outer circumferential surface of the photosensitive drum **111**. The exposure unit **113** exposes the outer circumferential surface of the photosensitive drum **111** to form an electrostatic latent image on the outer circumferential surface of the photosensitive drum **111**. Then, the development cartridge **200** supplies, as the development process, the toner to the outer circumferential surface of the photosensitive drum **111** to develop the electrostatic latent image into the toner image. The toner image on the outer circumferential surface of the photosensitive drum **111** is primarily transferred to an intermediate transfer belt **121** which will be described later. The cleaning device **114** removes the toner which is left on the outer circumferential surface of the photosensitive drum **111** without being transferred to the intermediate transfer belt **121**.

With reference back to FIG. **3**, the print unit **1000** includes a transfer belt unit **120**. The print unit **1000** also includes a secondary transfer roller **130**.

The transfer belt unit **120** is arranged below the image formation units **110**. The transfer belt unit **120** includes the intermediate transfer belt **121**. The intermediate transfer belt **121** is a seamless belt.

The intermediate transfer belt **121** is stretched with a plurality of rollers including a drive roller **122** (for the other rollers, symbols are omitted), and is supported rotatably. The intermediate transfer belt **121** is in contact with the outer circumferential surfaces of the photosensitive drums **111**, and is rotated in such a state. The drive roller **122** is rotated by transmission of power from a belt motor (not shown). The drive roller **122** is rotated, and thus the intermediate transfer belt **121** follows the rotation to rotate. The intermediate transfer belt **121** corresponds to an "image carrying member".

The transfer belt unit **120** includes primary transfer rollers **123**. The primary transfer rollers **123** are assigned to the individual colors of cyan, magenta, yellow and black, respectively. The primary transfer rollers **123** are arranged on the inner circumferential side of the intermediate transfer belt **121**. The primary transfer rollers **123** are respectively arranged opposite the photosensitive drums **111** which carry the toner images of the corresponding colors through the intermediate transfer belt **121**.

The secondary transfer roller **130** is arranged opposite the drive roller **122** through the intermediate transfer belt **121**. The secondary transfer roller **130** is pressed against the intermediate transfer belt **121**. The secondary transfer roller **130** forms the transfer nip between the intermediate transfer belt **121** and itself. The secondary transfer roller **130** corresponds to a "transfer roller".

Each of the image formation units **110** forms the toner image of the corresponding color. Then, each of the primary transfer rollers **123** primarily transfers the toner image to the outer circumferential surface of the intermediate transfer belt **121**. The intermediate transfer belt **121** is rotated while carrying, on the outer circumferential surface, the toner images primarily transferred from the photosensitive drums **111**. While the sheet S is being passed through the transfer nip, the sheet S makes contact with the outer circumferential surface of the intermediate transfer belt **121**. The secondary transfer roller **130** forms a transfer electric field between the intermediate transfer belt **121** and itself to secondarily transfer the toner images to the sheet S being passed through the transfer nip.

The print unit **1000** includes a fixing roller pair **140**. The fixing roller pair **140** includes a heating roller and a pressure roller. The heating roller incorporates a heater. The pressure roller is pressed against the heating roller to form a fixing nip between the heating roller and itself. The fixing roller pair **140** is rotated while nipping the sheet S conveyed from the transfer nip. In other words, the fixing roller pair **140** heats and pressurizes the sheet S being passed through the fixing nip. In this way, the fixing roller pair **140** fixes, to the sheet S, the toner images transferred to the sheet S. Thereafter, the sheet S is ejected to an ejection tray ET.

The image forming apparatus **100** can perform, as the print job, not only a single-sided print job for printing the toner images on only one side of the sheet S but also a double-sided print job for printing the toner images on both sides of the sheet S. In order to perform the double-sided print job, the image forming apparatus **100** includes the double-sided printing conveyance path DP.

The double-sided printing conveyance path DP branches from the main conveyance path MP on the downstream side of the fixing nip of the main conveyance path MP in a sheet conveyance direction. Then, the double-sided printing conveyance path DP merges into the main conveyance path MP on the upstream side of the transfer nip of the main conveyance path MP in the sheet conveyance direction.

When the performance job is the single-sided print job, the sheet S is passed through the transfer nip only once, and transfer processing is performed once on the sheet S being passed through the transfer nip. Then, after the first round of the transfer processing, the sheet S is ejected to the ejection tray ET without being processed.

When the performance job is the double-sided print job, the sheet S is passed through the transfer nip twice so that the transfer processing is performed once on each of the front and back surfaces of the sheet S. Specifically, when the sheet S is passed through the transfer nip for the first time, the transfer processing is performed on one side of the sheet S. After the first round of the transfer processing, the sheet S is switched back before the sheet S is fully ejected to the ejection tray ET after the back end of the sheet S is passed through the fixing nip. In this way, the sheet S is drawn into the double-sided printing conveyance path DP from the back end of the sheet S.

Thereafter, the sheet S is conveyed along the double-sided printing conveyance path DP. Then, the sheet S is returned to the main conveyance path MP on the upstream side of the transfer nip in the sheet conveyance direction. The sheet S returned to the main conveyance path MP is passed through the transfer nip again. Here, the directions of the front and back surfaces of the sheet S are reversed from the directions when the sheet S was previously passed through the transfer nip. In this way, when the sheet S is passed through the transfer nip for the second time, the transfer processing is performed on the other side opposite to the one side of the sheet S.

<Units in Apparatus>

The image forming apparatus **100** includes a plurality of units. Some of the units will be described below.

As shown in FIG. **5**, the image forming apparatus **100** includes a main body-side unit **1** and a conveyance unit **2**. The main body-side unit **1** corresponds to a "first unit", and the conveyance unit **2** corresponds to a "second unit". FIG. **5** shows a state where the rear cover CV is slightly opened (state where the conveyance unit **2** is separate from the main body-side unit **1**).

The main body-side unit **1** includes the transfer belt unit **120**. In other words, the main body-side unit **1** includes the intermediate transfer belt **121**.

The conveyance unit **2** mainly performs the conveyance of the sheet S. The conveyance unit **2** will be described below with reference to FIGS. **5** to **8**.

The conveyance unit **2** includes unit-side portions **20**. The unit-side portion **20** is arranged on each of one side and the other side in the width direction D**1**. The pair of unit-side portions **20** are opposite each other in the width direction D**1**.

The conveyance unit **2** includes conveyance guides G**1** which partition parts of the main conveyance path MP. The conveyance guides G**1** are located between the pair of unit-side portions **20** in the width direction D**1**. The conveyance guides G**1** partition parts of the main conveyance path MP together with conveyance guides (not shown) in the other units.

The conveyance unit **2** also includes conveyance guides G**2** which partition parts of the double-sided printing conveyance path DP. The conveyance guides G**2** are located between the pair of unit-side portions **20** in the width direction D**1**. The conveyance guides G**2** partition parts of the double-sided printing conveyance path DP together with a conveyance guide (symbol of which is omitted) provided in the rear cover CV.

Here, the conveyance unit **2** includes the secondary transfer roller **130**. However, the conveyance unit **2** is a separate unit from the main body-side unit **1**. In this configuration, the intermediate transfer belt **121** and the secondary transfer roller **130** form the transfer nip, and thus the conveyance unit **2** is located with respect to the main body-side unit **1**.

The conveyance unit **2** can be turned around an axis line extending in the width direction D**1**. For example, the main body frame includes a turning shaft RA**2** which extends in the width direction D**1**. The turning shaft RA**2** supports an end portion of the conveyance unit **2** on a lower side. The conveyance unit **2** is turned with the turning shaft RA**2** used as a support point such that an end portion (that is, an end portion on an upper side) on a side opposite to the side supported by the turning shaft RA**2** swings. The conveyance unit **2** is turned around the axis line of the turning shaft RA**2**, and thus the conveyance unit **2** can be displaced between a first position and a second position.

The first position is a position in which the conveyance unit **2** is located with respect to the main body-side unit **1**. The position of the conveyance unit **2** shown in FIG. **6** is the first position. The conveyance unit **2** is retained in the first position (that is, the conveyance unit **2** is located with respect to the main body-side unit **1**), and thus the secondary transfer roller **130** is pressed against the intermediate transfer belt **121**. In this way, the transfer nip is formed between the intermediate transfer belt **121** and the secondary transfer roller **130**.

The second position is a position other than the first position, and is a position in which the conveyance unit **2** is not located with respect to the main body-side unit **1**. In a state where the conveyance unit **2** is in the second position, it is impossible to locate the conveyance unit **2** with respect to the main body-side unit **1**. The position of the conveyance unit **2** shown in FIG. **5** is the second position.

In a state where the conveyance unit **2** is retained in the first position, a part of the main conveyance path MP is formed by the conveyance guides G**1** of the conveyance unit **2** and the conveyance guides of the other units. In a state where the conveyance unit **2** is retained in the first position, and the rear cover CV is closed, a part of the double-sided printing conveyance path DP is formed by the conveyance guides G**2** of the conveyance unit **2** and the conveyance guide (symbol of which is omitted) of the rear cover CV.

For example, when a jam occurs while the print job is being performed, a jam processing operation of opening the rear cover CV and turning the conveyance unit **2** to the second position is performed. Since the rear cover CV is opened, and thus the double-sided printing conveyance path DP is exposed, the sheet S jammed in the double-sided printing conveyance path DP can be removed. Since the conveyance unit **2** is turned to the second position, and thus the main conveyance path MP is exposed, the sheet S jammed in the main conveyance path MP can be removed.

After the jam processing operation, an operation of closing the rear cover CV is performed. For example, the rear cover CV is turned in a direction in which the rear cover CV is closed, and thus the conveyance unit **2** is pushed by the rear cover CV. In this way, in synchronization with the closing of the rear cover CV, the conveyance unit **2** is returned to the first position.

In this configuration, it is necessary to reliably retain the conveyance unit **2** in the first position during a normal operation (for example, while the print job is being performed). If the conveyance unit **2** is shifted from the first position while the print job is being performed, the image quality may be lowered or a jam may occur.

Hence, the image forming apparatus **100** includes a locating member **3**. The locating member **3** locates the conveyance unit **2** with respect to the main body-side unit **1**. In other words, the locating member **3** retains the conveyance unit **2** in the first position. Furthermore, in other words, the locating member **3** restricts the turning of the conveyance unit **2** from the first position to the second position.

<Locating of Unit>

The structure of the locating member **3** and a structure for attaching the locating member **3** to the conveyance unit **2** will be described below with reference to FIGS. **9** and **10**.

The locating member **3** is attached to each of the pair of unit-side portions **20** in the conveyance unit **2**. The structure of the locating member **3** and the structure for attaching the locating member **3** to the conveyance unit **2** are the same on each of one side and the other side. Hence, a description will be given below for only the one side, and the description for the other side is omitted because the following description can be used for the description.

The locating member **3** is arranged outward of the unit-side portion **20** in the width direction D**1**. The secondary transfer roller **130** and the like are arranged inward of the unit-side portion **20** in the width direction D**1**. The "outward" in the width direction D**1** refers to a direction which extends from the center of the conveyance unit **2** in the width direction D**1** toward one side or the other side in the width direction D**1**. The "inward" in the width direction D**1** refers to a direction which extends from the one side or the other side of the conveyance unit **2** in the width direction D**1** toward the center in the width direction D**1**.

The unit-side portion **20** includes a support shaft **21** which supports the locating member **3** outward in the width direction D**1**. The support shaft **21** protrudes outward from the unit-side portion **20** in the shape of a cylinder or a tube. The outer shape of the support shaft **21** is circular when viewed in the width direction D**1**. For example, the unit-side portion **20** is made of resin. The support shaft **21** is integrally molded with the unit-side portion **20**.

The locating member **3** includes a cylindrical portion **30**. In a state where the locating member **3** is attached to the unit-side portion **20**, the width direction D**1** is the direction of the cylindrical shaft of the cylindrical portion **30**. The support shaft **21** is inserted into the cylindrical portion **30**, and thus the locating member **3** is attached to the unit-side portion **20**. The cylindrical portion **30** is fitted to the support shaft **21** from outside in the width direction D**1**. In the following description, a circumferential direction around the center of the cylindrical shaft of the cylindrical portion **30** fitted to the support shaft **21** (that is, the support shaft **21**) when viewed in the width direction D**1** is simply referred to as a circumferential direction.

The cylindrical portion **30** can slide in the circumferential direction with respect to the outer circumferential surface of the support shaft **21**. In other words, the support shaft **21** supports the locating member **3** such that the locating member **3** can be displaced in the circumferential direction. The cylindrical portion **30** can also slide in the width direction D**1** with respect to the outer circumferential surface of the support shaft **21**.

The locating member **3** includes a hook portion **31**. The hook portion **31** is arranged radially outward of the cylindrical portion **30** when viewed in the width direction D**1**. The hook portion **31** is a part which extends radially outward of the cylindrical portion **30** (that is, in a direction away from the cylindrical portion **30**) when viewed in the width direction D**1**, and is a hook-shaped part which can engage with a boss portion **10** to be described later from one side in the circumferential direction. The cylindrical portion **30** is displaced in the circumferential direction, and thus the hook portion **31** is turned around the center of the cylindrical shaft of the cylindrical portion **30**.

In order to use the locating member **3** to retain the conveyance unit **2** in the first position, the main body-side unit **1** includes the boss portion **10** (see FIGS. **5** and **6**) which protrudes in the width direction D**1**. The boss portion **10** is arranged on the turning path of the hook portion **31** when the hook portion **31** is turned around the center of the cylindrical shaft of the cylindrical portion **30** as viewed in the width direction D**1** (that is, when the hook portion **31** is turned in the circumferential direction).

When the conveyance unit **2** is in the first position, the hook portion **31** engages with the boss portion **10** so as to hold the boss portion **10** from the one side in the circumferential direction. In this way, the hook portion **31** restricts the turning of the conveyance unit **2** from the first position to the second position. Consequently, the conveyance unit **2** is retained in the first position. In other words, the conveyance unit **2** is located with respect to the main body-side unit **1**.

However, the locating member **3** is not fixed to the unit-side portion **20** (that is, the conveyance unit **2**). In other words, the locating member **3** is removable with respect to the unit-side portion **20**. In this configuration, if some countermeasure is not taken, the cylindrical portion **30** may come off the support shaft **21** such that the locating member **3** drops off the unit-side portion **20**.

If the locating member **3** drops off the unit-side portion **20**, the engagement of the hook portion **31** with the boss portion **10** is released, and thus the conveyance unit **2** is turned from the first position to the second position. In other words, the conveyance unit **2** is shifted with respect to the main body-side unit **1**.

In order to suppress such inconvenience, the unit-side portion **20** includes a restriction portion **23**. The locating member **3** includes a portion to be restricted **32**.

The portion to be restricted **32** is arranged radially outward of the cylindrical portion **30** and in a position different from the hook portion **31** in the circumferential direction when viewed in the width direction D**1**. For example, the locating member **3** includes a part **32*a*** which extends radially outward of the cylindrical portion **30** when viewed in the width direction D**1**, and includes, as the portion to be restricted **32**, a protrusion part which protrudes from the part **32*a*** to the one side in the circumferential direction. The cylindrical portion **30** can be displaced in the circumferential direction. Hence, the portion to be restricted **32** arranged radially outward of the cylindrical portion **30** can be turned around the center of the cylindrical shaft of the cylindrical portion **30**.

The restriction portion **23** is arranged in a predetermined position (which is hereinafter referred to as a restriction position) on the turning path of the portion to be restricted **32** which is turned around the center of the cylindrical shaft of the cylindrical portion **30** when viewed in the width direction D**1**. The hook portion **31** is caused to engage with the boss portion **10** from the one side in the circumferential direction, and thus the portion to be restricted **32** is arranged in the restriction position. In this way, in a state where the hook portion **31** engages with the boss portion **10** from the one side in the circumferential direction, the restriction portion **23** overlaps the portion to be restricted **32** from the outside in the width direction D**1**.

In a state where the restriction portion **23** overlaps the portion to be restricted **32** from the outside in the width direction D**1**, even when the portion to be restricted **32** (that is, the locating member **3**) attempts to move outward in the width direction D**1**, the portion to be restricted **32** collides with the restriction portion **23**. In other words, the restriction portion **23** restricts the movement of the locating member **3** outward in the width direction D**1**. Consequently, it is possible to suppress the dropping of the locating member **3** off the unit-side portion **20**.

However, when the locating member **3** is displaced to the other side in the circumferential direction, the restriction portion **23** and the portion to be restricted **32** do not overlap each other in the width direction D**1**. In this case, the locating member **3** inconveniently drops off the unit-side portion **20**.

Hence, the image forming apparatus **100** further includes a torsion coil spring **4**. The torsion coil spring **4** is attached to the locating member **3**. The torsion coil spring **4** uses its biasing force to restrict the displacement of the locating member **3** to the other side in the circumferential direction.

The torsion coil spring **4** includes a winding spring portion **40** (that is, a coil-shaped part). The winding spring portion **40** is fitted to the cylindrical portion **30** to be arranged on the outer circumference of the cylindrical portion **30**. In other words, the cylindrical portion **30** is inserted into the winding spring portion **40**. The winding spring portion **40** is fitted to the cylindrical portion **30**, and thus the torsion coil spring **4** is attached to the locating member **3**.

The torsion coil spring **4** includes a first arm portion **41** which extends from one end of the winding spring portion **40** and a second arm portion **42** which extends from the other end of the winding spring portion **40**. Each of the first arm portion **41** and the second arm portion **42** extends tangentially to the winding spring portion **40** when viewed in the width direction D**1**.

The first arm portion **41** engages with the locating member **3**. Specifically, the locating member **3** includes a first engagement portion **33**. The first engagement portion **33** is arranged radially outward of the cylindrical portion **30** when viewed in the width direction D**1**. The first arm portion **41** engages with the first engagement portion **33**.

The second arm portion **42** engages with the conveyance unit **2**. Specifically, a second engagement portion **22** is provided in the unit-side portion **20**. An edge portion of an opening **20*a*** which penetrates the unit-side portion **20** in the width direction D**1** functions as the second engagement portion **22**. The second arm portion **42** engages with the second engagement portion **22**.

The first arm portion **41** engages with the first engagement portion **33** (that is, the locating member **3**), the second arm portion **42** engages with the second engagement portion **22** (that is, the unit-side portion **20**) and thus the torsion coil spring **4** generates a force for biasing the locating member **3** to the one side in the circumferential direction. In other words, the torsion coil spring **4** biases the locating member **3** in a direction which causes the hook portion **31** to engage with the boss portion **10** (that is, to the one side in the circumferential direction). Furthermore, in other words, the torsion coil spring **4** restricts the displacement of the locating member **3** to the other side in the circumferential direction.

In the present embodiment, it is possible to suppress the displacement of the locating member **3** to the other side in the circumferential direction. When the locating member **3** is not displaced to the other side in the circumferential direction, the overlapping of the restriction portion **23** and the portion to be restricted **32** in the width direction D**1** is maintained, and thus it is possible to suppress the dropping of the locating member **3** off the unit-side portion **20**. In this way, it is possible to suppress the turning of the conveyance unit **2** from the first position to the second position. In other words, it is possible to suppress the shifting of the conveyance unit **2** with respect to the main body-side unit **1**.

In the present embodiment, the torsion coil spring **4** is used in order to restrict the turning of the conveyance unit **2** from the first position to the second position. The torsion coil spring **4** is used, and thus it is possible to reliably restrict, with the locating member **3**, the turning of the conveyance unit **2** (that is, the shifting of the conveyance unit **2**) without fixing the locating member **3** to the unit-side portion **20**. In other words, it is possible to suppress the dropping of the locating member **3** off the unit-side portion **20** caused by the displacement of the locating member **3** in the circumferential direction without fixing the locating member **3** to the unit-side portion **20**. When the locating member **3** is not fixed to the unit-side portion **20**, it is easy to disassemble the locating member **3** and an area therearound (for example, to remove the locating member **3** from the unit-side portion **20**).

In the present embodiment, in the configuration in which the turning of the conveyance unit **2** (that is, the shifting thereof) is restricted with the locating member **3**, it is possible to suppress the shifting of the conveyance unit **2** while suppressing a decrease in the disassembly of the locating member **3** and the area therearound. An operation of removing the locating member **3** from the conveyance unit **2** is only to release the engagement of the second arm portion **42** of the torsion coil spring **4** with the second engagement portion **22** of the unit-side portion **20** and to turn the locating member **3** to the other side in the circumferential direction.

<Attachment of Locating Member>

The locating member **3** includes a temporary fixing portion **34**. The temporary fixing portion **34** can engage with the second arm portion **42** of the torsion coil spring **4**. The temporary fixing portion **34** is used when the locating member **3** is attached to the unit-side portion **20** (that is, the conveyance unit **2**).

A step of attaching the locating member **3** to the unit-side portion **20** will be described below with reference to FIGS. **11** and **12**.

The torsion coil spring **4** is first attached to the locating member **3** before the locating member **3** is attached to the unit-side portion **20**. Here, the winding spring portion **40** is fitted to the cylindrical portion **30**, and the first arm portion **41** is caused to engage with the first engagement portion **33**. Furthermore, the second arm portion **42** is caused to engage with the temporary fixing portion **34**. In other words, the torsion coil spring **4** is temporarily fixed to the locating member **3**.

Then, the cylindrical portion **30** is fitted to the support shaft **21**. Here, the portion to be restricted **32** is retained so as not to overlap the restriction portion **23** in the width direction D1, and in this state, the cylindrical portion **30** is fitted to the support shaft **21**. In this way, a state shown in FIG. **11** is achieved. Until the fitting of the cylindrical portion **30** to the support shaft **21** is completed, the engagement of the second arm portion **42** with the temporary fixing portion **34** is maintained.

After the cylindrical portion **30** is fitted to the support shaft **21**, the engagement of the second arm portion **42** is switched from the temporary fixing portion **34** to the second engagement portion **22** (that is, the unit-side portion **20**). Here, an operator grasps the second arm portion **42**, removes the second arm portion **42** from the temporary fixing portion **34** and causes the second arm portion **42** to engage with the second engagement portion **22**.

The winding spring portion **40** is fitted to the cylindrical portion **30**, and in a state where the first arm portion **41** engages with the first engagement portion **33**, the second arm portion **42** is caused to engage with the second engagement portion **22**, with the result that the force for biasing the locating member **3** to the one side in the circumferential direction is generated in the torsion coil spring **4**. Consequently, a state shown in FIG. **12**, that is, a state where the portion to be restricted **32** and the restriction portion **23** overlap each other in the width direction D1 is retained.

In the present embodiment, the temporary fixing portion **34** is provided in the locating member **3**, and thus workability when the locating member **3** is attached to the unit-side portion **20** is enhanced. Specifically, the winding spring portion **40** is fitted to the cylindrical portion **30**, the first arm portion **41** is caused to engage with the first engagement portion **33** and the second arm portion **42** is caused to engage with the temporary fixing portion **34**, with the result that a state where the torsion coil spring **4** is biased in the direction of torsion is retained. In this way, during the operation of attaching the locating member **3** to the unit-side portion **20** (specifically, before the cylindrical portion **30** is fitted to the support shaft **21**), it is possible to suppress the dropping of the torsion coil spring **4** off the locating member **3**. This is convenient for the operator because during the operation of attaching the locating member **3** to the unit-side portion **20**, it is not necessary to pay attention to the dropping of the torsion coil spring **4**.

As a variation, the following configuration may be adopted. The variation will be described below with reference to FIG. **13**.

In the variation, the conveyance unit **2** includes a guide portion **24**. The guide portion **24** is arranged inward of the unit-side portion **20** in the width direction D**1**. The guide portion **24** includes an inclination surface **240** which is inclined upward from outside to inside in the width direction D**1**.

The inclination surface **240** is arranged on the movement path of the tip of the second arm portion **42** when the cylindrical portion **30** is fitted to the support shaft **21** (specifically, the tip of the second arm portion **42** on a side opposite to the side of the connection to the winding spring portion **40**). Specifically, when the cylindrical portion **30** is fitted to the support shaft **21** in a state where the second arm portion **42** engages with the temporary fixing portion **34**, the tip of the second arm portion **42** reaches the inclination surface **240** of the guide portion **24** arranged inward of the unit-side portion **20** in the width direction D1 from the outside of the unit-side portion **20** in the width direction D1 through the second engagement portion **22** (that is, the opening **20***a* which penetrates the unit-side portion **20** in the width direction D1). In other words, the second arm portion **42** makes contact with the inclination surface **240**. Thereafter, the second arm portion **42** is moved along the inclination surface **240**.

The second arm portion **42** is moved along the inclination surface **240**, and thus the position of the tip of the second arm portion **42** in the up/down direction is displaced upward. In other words, the second arm portion **42** is displaced upward beyond a position which allows engagement with the temporary fixing portion **34**. In this way, the engagement of the second arm portion **42** with the temporary fixing portion **34** is released. Then, the second arm portion **42** released from the temporary fixing portion **34** engages with the second engagement portion **22** (that is, the edge portion of the opening **20***a* of the unit-side portion **20**).

In the variation, the cylindrical portion **30** is fitted to the support shaft **21**, and thus the engagement of the second arm portion **42** is switched from the temporary fixing portion **34** to the second engagement portion **22** without the operation of switching the engagement of the second arm portion **42** from the temporary fixing portion **34** to the second engagement portion **22** being performed. Since the operation of switching the engagement of the second arm portion **42** from the temporary fixing portion **34** to the second engagement portion **22** is not performed additionally, the convenience of the operator is enhanced,

It should be considered that the embodiment disclosed herein is illustrative in all respects, and not restrictive. The scope of the present disclosure is indicated not by the description of the above embodiment but by the scope of claims, and furthermore, meanings equivalent to the scope of claims and all changes in the scope are included therein.

What is claimed is:

1. A unit locating mechanism comprising:

a first unit that includes a boss portion;

a second unit that can be displaced between a first position located with respect to the first unit and a second position other than the first position by being turned around an axis line extending in a predetermined direction; and

a locating member that retains the second unit in the first position,

wherein the locating member includes:

a cylindrical portion in which the predetermined direction is a direction of a cylindrical shaft;

a hook portion that

is arranged radially outward of the cylindrical portion when viewed in the predetermined direction and

engages with the boss portion so as to hold the boss portion from one side in a circumferential direction around a center of the cylindrical shaft of the cylindrical portion; and

a portion to be restricted that is arranged radially outward of the cylindrical portion and in a position different from the hook portion in the circumferential direction when viewed in the predetermined direction,

the second unit includes:

a unit-side portion;

a support shaft

which protrudes outward in the predetermined direction from the unit-side portion,

to which the cylindrical portion is fitted from outside in the predetermined direction and

which supports the locating member such that the locating member can be displaced in the circumferential direction; and

a restriction portion that is arranged on a turning path of the portion to be restricted which is turned around the center of the cylindrical shaft of the cylindrical portion when viewed in the predetermined direction,

the hook portion engages with the boss portion when the second unit is in the first position to restrict the turning of the second unit from the first position to the second position,

the restriction portion overlaps, in a state where the hook portion engages with the boss portion, the portion to be restricted from the outside in the predetermined direction so as to restrict movement of the locating member outward in the predetermined direction,

the unit locating mechanism further includes a torsion coil spring that includes

a winding spring portion which is arranged on the cylindrical portion,

a first arm portion which extends from one end of the winding spring portion and

a second arm portion which extends from an other end of the winding spring portion and

the first arm portion engages with the locating member, and the second arm portion engages with the unit-side portion such that the torsion coil spring biases the locating member in a direction which causes the hook portion to engage with the boss portion.

2. The unit locating mechanism according to claim **1**,

wherein the locating member includes a temporary fixing portion that can engage with the second arm portion.

3. The unit locating mechanism according to claim **2**,

wherein the unit-side portion includes, as an engagement portion with which the second arm portion engages, an edge portion of an opening that penetrates in the predetermined direction,

the second unit includes an inclination surface that

is arranged inward of the unit-side portion in the predetermined direction and

is inclined upward from the outside to inside in the predetermined direction and

when the cylindrical portion is fitted to the support shaft in a state where the second arm portion engages with the temporary fixing portion, the second arm portion reaches the inclination surface through the opening,

moves along the inclination surface to release the engagement of the second arm with the temporary fixing portion and

engages with the edge portion of the opening.

4. An image forming apparatus comprising:

the unit locating mechanism according to claim **1**,

wherein the first unit is a unit that includes an image carrying member which carries an image,

the second unit is a unit that includes a transfer roller which is pressed against the image carrying member and

a sheet is conveyed into a transfer nip formed between the image carrying member and the transfer roller, and the image is transferred to the sheet.

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